

EXECUTIVE SUMMARY

A prototype we can demonstrate

We are asking for support to turn our digital air-cleaning studies into one safely built and measured indoor prototype.

Vision for 2026

Build and test one air-cleaning prototype. Use its results to decide whether a larger outdoor tower is worth developing. Outdoor coverage is not proven.

Work completed

Airflow studies, filter-bypass investigation, cylindrical packaging studies, service-access checks and measurement software. This is digital development; physical prototypes built: 0.

Key decisions

Start with a fan, coarse filter and sealed HEPA filter. Use standard bought parts and a local workshop. Keep solar, water and gas-treatment experiments out of the first demonstration.

Unresolved challenges

Choose a fan that can handle filter resistance; confirm real parts, seals, supports and safe wiring. Then measure actual airflow, particle reduction, power and noise.

Support requirements

Fund the first component selection and reviewed build, instruments and paid testing. College and NGO support is financial; no college lab is assumed.

Plan for the upcoming month

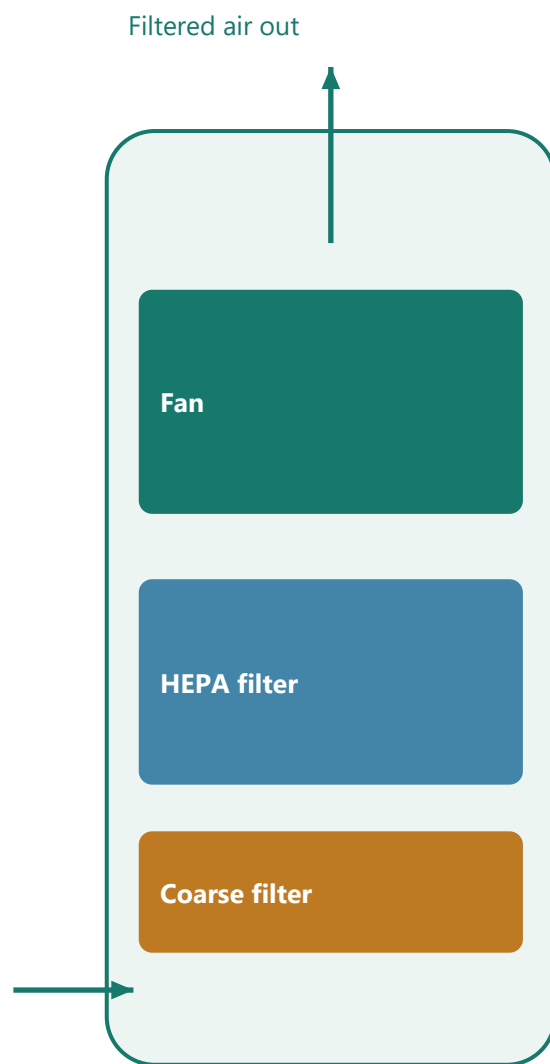
Obtain component and workshop quotations, select a demonstration route, complete reviewed part-specific drawings and arrange assembly/testing. Dates depend on confirmed parts and service availability.

Decision requested: support one bounded indoor prototype and its measurements, not an unverified city-scale clean-air claim.

THE PROPOSED PROTOTYPE

How the tower would work

A cylindrical appearance is an option. The first working core must filter air safely before we spend on the finished outer shell.



Illustrative layout only

1 Draw room air in

A guard keeps people away from moving parts.

2 Catch larger dust

A removable coarse filter protects the next stage.

3 Filter fine particles

A HEPA element needs a sealed frame so air cannot go around it.

4 Return filtered air

An OEM fan pulls air through the filters. Its actual delivered flow must be measured.

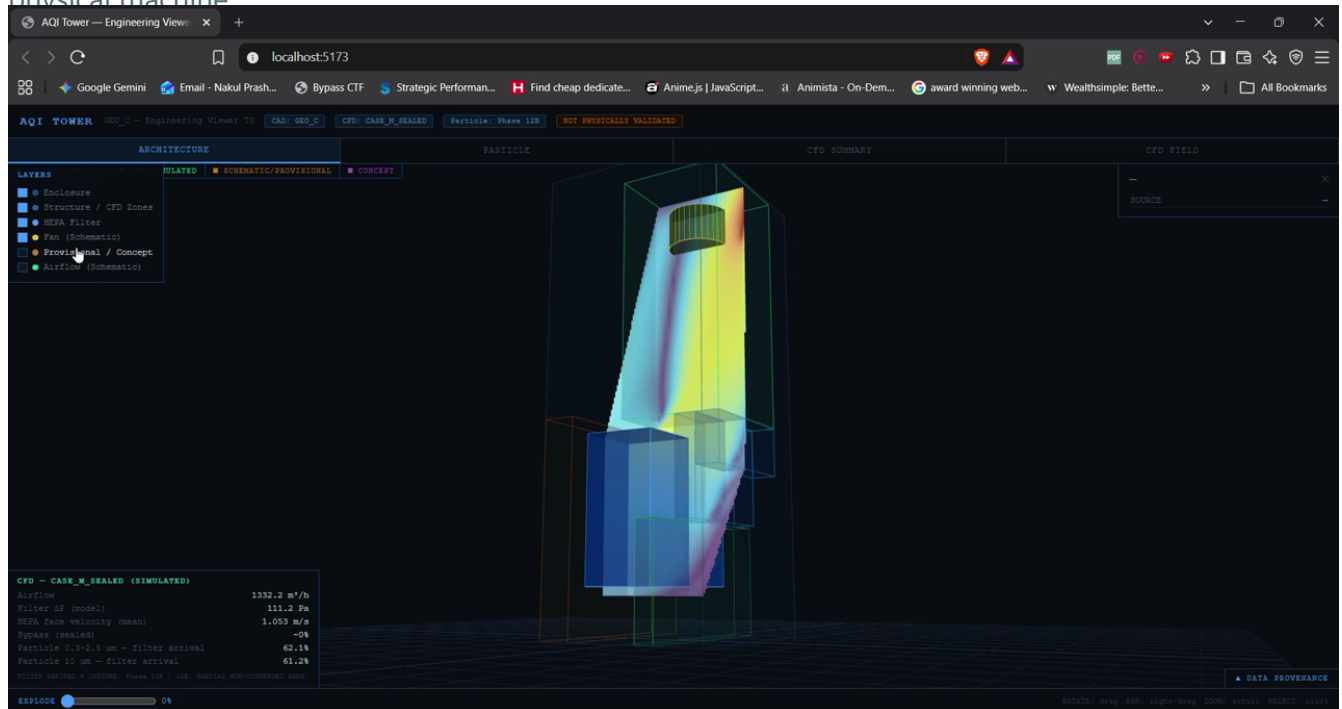
The structure, guards, service hatch and electrical enclosure are part of the prototype, not optional decoration. The drawing shows functions, not final sizes or wiring.

What we will measure: airflow, filter pressure, room particle trends, electrical power, noise and installed filter sealing. No virus-removal or health-protection claim is proposed.

ACTUAL PROJECT PROGRESS

What already exists

The project has digital evidence to show now. It does not yet have measured performance from a physical machine



Existing project viewer: historical digital geometry and airflow study

Airflow and filters

We investigated the air path and a modeled leak around the HEPA filter. A historical simulation predicted about 1,332 m³/h; it was partial and not fully converged. This is not a tested tower rating.

Packaging and maintenance

FreeCAD studies explored a cylinder, actual-depth filter envelopes, supports and a removable service hatch. Nominal fit checks do not establish structural strength or leak-tightness.

Monitoring and build preparation

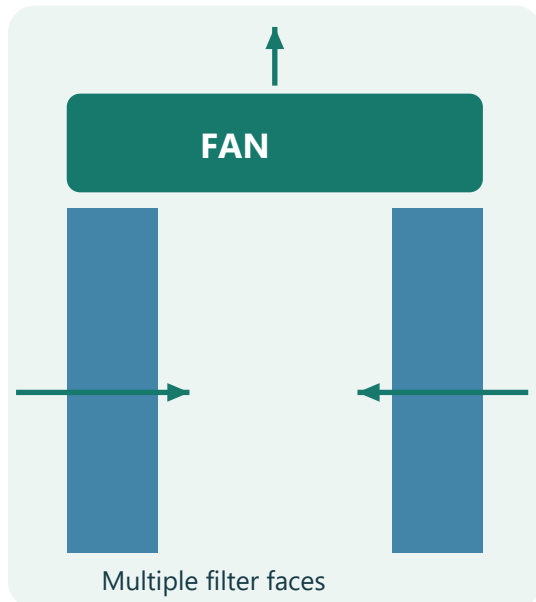
Offline recording and analysis tools are available. Functional start/stop plans and supplier questions are prepared. No sensors connected, supplier order or physical commissioning is recorded.

GEO_C is simply the internal name of an airflow layout tested on the computer. It is not a product name, a manufactured tower or a safety approval.

REAL ONLINE REFERENCES

We can learn from existing designs

We can reuse documented ideas and standard components, while keeping outside results separate from our own evidence.



Original diagram of a published principle

Public construction precedent

EPA describes and tested DIY indoor air cleaners with fans and MERV 13 filters, including multi-filter arrangements. These give us a concrete example of fan-and-filter construction. MERV 13 is not HEPA.

The EPA fan safety guidance is specific to its tested designs and US equipment. It does not approve an arbitrary Indian 230 V fan or our tower. Local compatibility and qualified review remain necessary.

Real component references

S&P TD-SILENT ECOWATT: published fan curves and drawings; our comparison candidate is not selected and is not confirmed adequate at the assumed loaded duty.

Sensirion SEK-SPS30: a published particle-sensor evaluation kit candidate for logging. Actual kit, Windows setup, calibration/reference checks and supply remain to be confirmed.

Our originality must be assessed separately. Referencing a public design does not make the entire project patentable, nor transfer its test results to our machine.

EPA public construction and research reference

A FASTER SHOWABLE MILESTONE

Demonstrate first and build honestly

Two different routes can help us move. A purchased reference appliance must never be presented as our own completed tower.

Route A A reference demonstration

Optional scope for team approval. Use an unmodified, locally supported commercial indoor purifier beside the project display and particle logging. This demonstrates filtration and our measurement workflow without designing a new mains appliance for the event. A Philips AC3059/65 India product page is one real reference; stock, price and suitability are unconfirmed.

What Route A delivers

A working reference appliance, instrument setup and recorded demonstration data, if arranged and tested. Label all results with the exact appliance used. This is not our tower prototype, proof of our CAD or an outdoor demonstration.

Route B Our custom fan and filter core

Build our guarded, sealed core using OEM parts and a reviewed workshop assembly. It preserves our tower direction, but still needs a selected fan/filter, safe supports/seals and rated electrical design. Current study CAD is not a cutting or wiring instruction.

Why we cannot order the old fan blindly

At a study flow of 1,300 m³/h, loaded-filter assumptions require about 326–413 Pa of static pressure. The S&P comparison chart estimate is about 230 Pa. These are unmeasured scenarios, but enough to show that selection needs attention.

What can be shown immediately

This report, the offline clickable prototype explainer and the existing digital viewer. A non-operating physical appearance mock-up is another optional milestone. None should be labelled a physically validated purifier.

Time: no supplier or workshop date is booked. An off-the-shelf reference demo avoids custom fabrication, but delivery and logging still need confirmation. The custom tower has a separate reviewed build schedule.

SUPPORT AND NEXT ACTION

What funding will unlock

We need a small, accountable first build stage rather than a promise of a finished outdoor network.

Support requirements

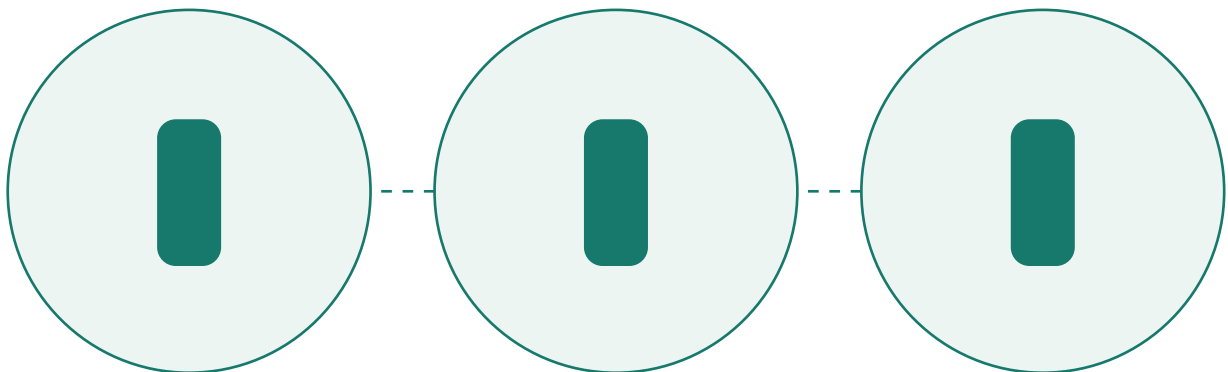
Fund component quotations and qualified design review; then approve selected parts, workshop assembly and paid commissioning/testing. Obtain particle monitoring and arrange airflow, power, noise and filter-integrity checks. Costs are UNKNOWN until quoted.

Plan for the upcoming month

Choose reference demonstration, custom core or both as separately labelled scopes. Confirm a usable indoor room and supply. Obtain fan/filter and service quotes. Revise drawings against real parts, review them and build only the released scope. Record actual results without creating smoke or hazardous pollutant challenges.

The longer-term outdoor vision

One tower may be investigated as a local cleaner-air zone. Several towers could later form a network. Wind, surrounding pollution and placement must be tested; there is no proven bubble size, cleaning percentage or tower spacing.



Future local-zone network concept only - circles are not measured coverage

Public references checked 3 October 2026

EPA fan-and-filter construction research

Philips AC3059/65 India reference appliance

S&P published fan catalogue - existing local archive

Sensirion SEK-SPS30 sensor kit - earlier OEM check

Local evidence: project status, CAD, historical CFD and measurement tools. Fan/kit web refreshes failed; earlier checked references are used. No purchase or supplier commitment is implied.